

Question number	Answer	Additional guidance	Mark
3(a)(i)	An answer that includes the following points		Graduate (2)
	Label	Part	
	L	dendrites	
	M	cell body / centron	
	N	axon	
	O	(pre)synaptic {endings /knob/bulb}	
		all correct 2 marks 1 or 2 or 3 correct 1 mark 0 correct 0 marks accept dendrite / dendron accept axon terminal do not accept synapse(s)	

Question number	Answer	Additional guidance	Mark
3(a)(ii)	D is the correct answer • A in not correct because lidocaine inhibits sodium ion channels • B in not correct because lidocaine inhibiting sodium ion channels • C in not correct because lidocaine inhibiting sodium ion channels		Computer (1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	(scanning) electron (microscope) /sem / em (1)	Do not accept electric / electrical /electronic Do not accept transmission electron microscope / tem Do not accept microgram	Graduate (1)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A calculation showing the following steps - correct measurement of X-Y in μm (1) - calculation of thickness of connecting tissue (1) OR - calculation of thickness (1) - conversion to μm (1)	11 000 μm $(11\ 000 \div 3300) = 3.3 / 3.33 / 3.333 / 3\ \mu\text{m}$ 1 mark for power of 10 errors e.g. 0.33 or 0.0033 or 333 $(11 \div 3300) = 0.00333$ $(1.1 \div 3300) = 0.0333$ $(0.00333) \times 1000 = 3.3 / 3.33 / 3.333 / 3\ \mu\text{m}$ 1 mark for power of 10 errors e.g. 0.33 or 0.0033 or 333 No ECF	Expert (2)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that includes three of the following points • myelin (sheath) acts as an insulator (1) • gaps (in myelin sheath) are called nodes of Ranvier /nodes of Ranvier have higher number of (voltage gated) sodium channels (1) • {action potentials/depolarization} can only occur at nodes on Ranvier (in myelinated neurones)(1) • (therefore) impulse jumps from one node of Ranvier to the next(1)	accept {action potentials / depolarization}occur along the whole unmyelinated neurone mp4 accept converse accept reference to saltatory conduction	Expert (3)